

simTIM

A Temporal Cybersecurity Simulator

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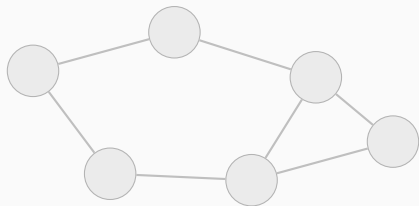
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Why model cybersecurity?

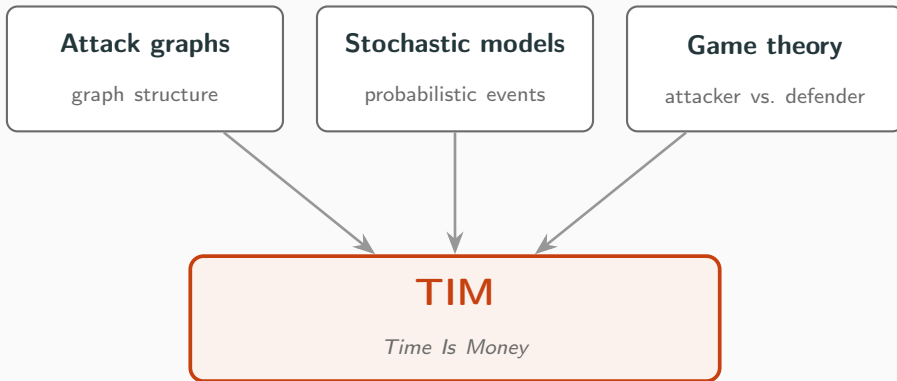


Which servers are vulnerable and need hardening?

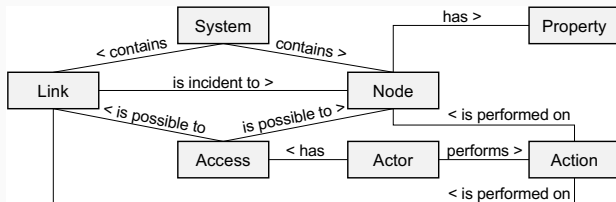
How should we change the network topology?

Simulations provide **what-if analysis**.

TIM: three families of security models



What TIM models



Nodes and links

servers, routers, connections

Attackers: maximise gain

Defenders: minimise damage

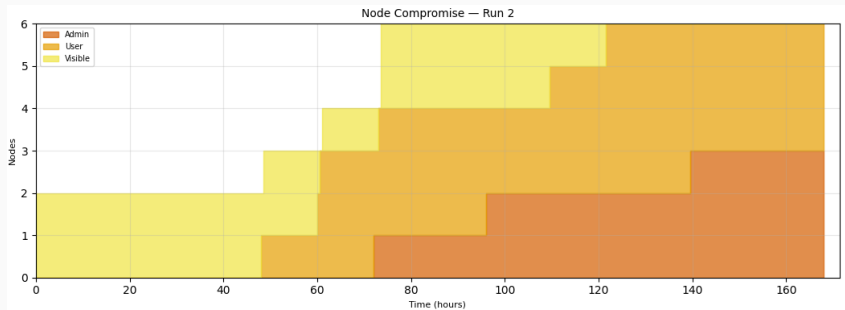
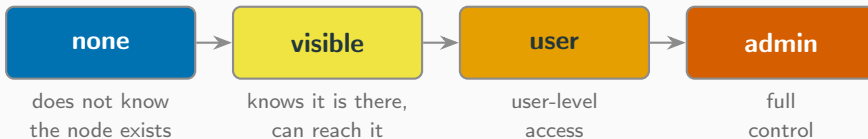
Actions

$$a = (\varphi_a, \psi_a, c_a, d_a, p_a)$$

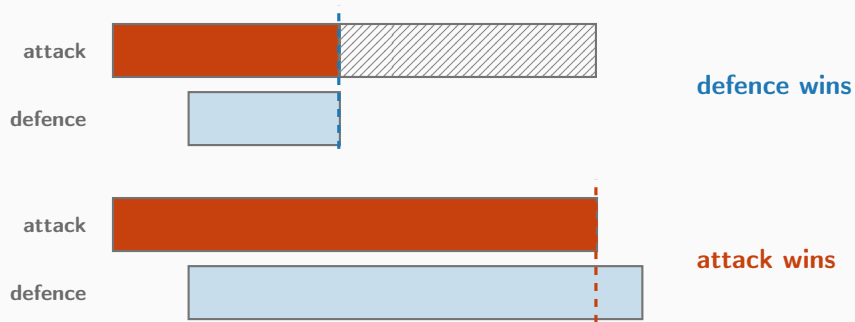
precondition, postcondition,

cost, duration, success probability

Access levels



Physical time

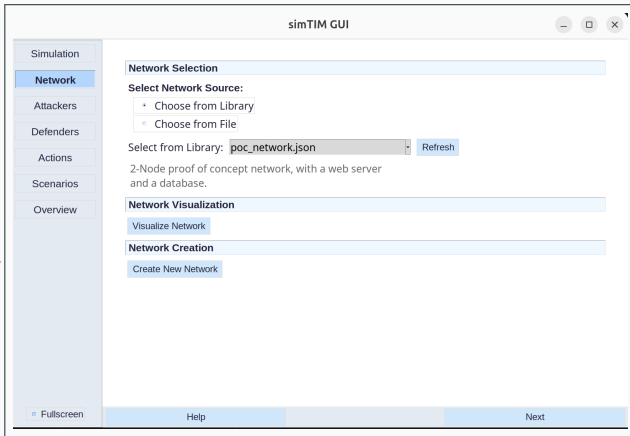


The outcome depends on **which one finishes first**.

simTIM: a full implementation of TIM

```
n1 = Node("web server")
n1.set("framework", ...)
n2 = Node("database")
...
attacker.do(a_tapestry, n1)
```

proof of concept
a *subset* of TIM



- ▶ graphical interface
- ▶ configurable simulations
- ▶ Monte Carlo runs
- ▶ visualisation

Demonstration

Open source and extensible



github.com/noahbehrisch/simTIM

Apache 2.0 · **actively developed**

```
make install  
make run
```

Four places to plug in:

- ▶ networks (JSON)
- ▶ actions (JSON)
- ▶ **strategies** (one class)
- ▶ detection engines (one class)